A line fitted to ten points: a toy paper demonstrating a reproducible workflow

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Abstract

Fitting lines to points attracts wide interest in ecology. Etcetera.

1 Introduction

At the core of ecology is the study of relationships between organisms and their environments. If you don't believe that, check out (Ghosh *et al.*, 2020). Ghosh *et al.* (2020) say some really convincing stuff. If you want to skip the parenthesis, use "citealp", like this: Ghosh *et al.*, 2020.

To illustrate, the ten points in Fig. 1 appear to lie close to a straight line, and the line shown in Fig. 1 is the one we fitted to them.

Letting $N_i(t)$ denote the adult population density of species $i = 1, 2$ at time t, and defining $N = N_1(t) + N_2(t)$, model equations are

$$N_i(t+1) = (1-\delta)N_i(t) + \delta N \frac{B_i(t)N_i(t)}{B_1(t)N_1(t) + B_2(t)N_2(t)} \quad (1)$$

for $i = 1, 2$. Here, δ is an adult mortality rate, and $B_i(t)$ is the fecundity of species i at time t. The model postulates that adults die at rate δ at each time step, and are replaced by juveniles in proportion to the reproductive outputs of the species that year. Note that $N = N_1(t) + N_2(t)$ is constant. This model has nothing to do with the rest of this toy paper; it is here to show what an equation looks like, and how to refer to one by number (it is equation 1).

2 Theory

We here pursue goal G1: to fit a line to ten points. Go back and read the Introduction, which is section 1, if you are confused.

3 Methods

The model is

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \tag{2}$$

for $i = 1, \ldots, 10$, where $x_i = i$, the y_i are the responses, β_0 is the intercept, β_1 is the slope, and the ϵ_i are errors. The "align" environment lines up several equations at the & signs, like this:

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}, \tag{3}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}. \tag{4}$$

These are the ordinary least squares estimates of the parameters of equation 2.

See SI section **??** for more details, and SI Fig. **??** for the same figure again.

4 Results

Our main result is that the fitted slope was $\hat{\beta}_1 = 1.959$ and the fitted intercept was $\hat{\beta}_0 = 0.367$ (Fig. 1). Written out in words, the slope was 1.959 and the intercept was 0.367, and if these look odd, remember the data were made up.

5 Discussion

5.1 Discusion topic one

Blah blah blah

5.2 Discusion topic two

More blah blah blah

6 Author contributions

Both authors contributed about equally to all parts of the paper.

7 Acknowledgments

The authors thank some people.

8 Funding information

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References

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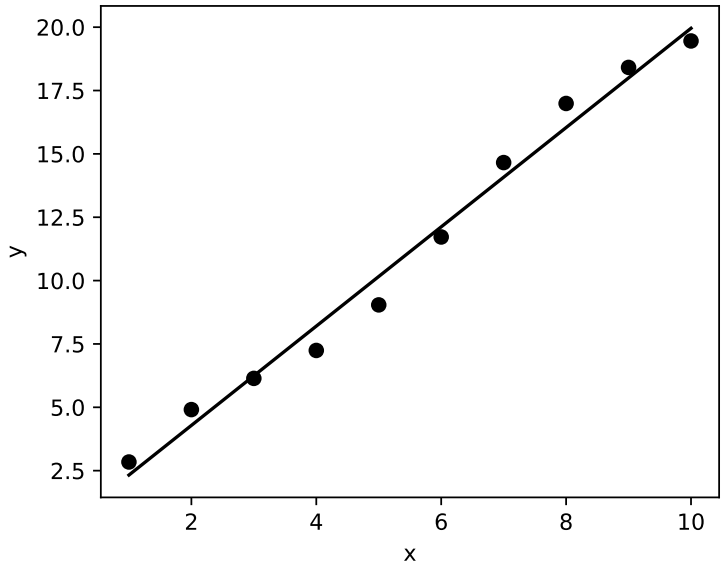


Figure 1: The ten toy data points (dots) and the line fitted to them by ordinary least squares (solid line). The fitted slope was 1.959 and the fitted intercept was 0.367; see section 4. Etcetera.